

Enhancing Healthcare Resource Allocation in England via Data Envelopment Analysis and Behavioural Sciences

Table of Contents

<i>Abstract</i>	<i>3</i>
<i>1. Introduction.....</i>	<i>3</i>
<i>1.1 Background on the NHS</i>	<i>3</i>
<i>1.2 Current Analytical Approaches</i>	<i>4</i>
<i>1.3 Limitations of Existing Methods</i>	<i>4</i>
<i>1.4 Model Selection</i>	<i>5</i>
<i>1.5 Overview</i>	<i>6</i>
<i>2. Literature Review.....</i>	<i>6</i>
<i>3. Research Methodology :</i>	<i>10</i>
<i>3.1 Model Evaluation.....</i>	<i>10</i>
<i>3.2 Data Source</i>	<i>12</i>
<i>3.3 Data Processing.....</i>	<i>13</i>
<i>4. Analysis & Discussions.....</i>	<i>14</i>
<i>4.1 Model Results.....</i>	<i>15</i>
4.1.1 Measuring Efficiency and Benchmarking	15
4.1.2 Visualizing Performance and Identifying Extremes.....	17
4.1.3 Connecting Financial Sustainability: Profit/Loss Analysis	18
4.1.4 Multi-Criteria Decision Analysis	20
4.1.5 Sensitivity and Scenario Analysis: Weighted DEA.....	22
<i>4.2 Actionable Recommendations.....</i>	<i>24</i>
<i>4.3 Summary and Impact</i>	<i>27</i>
<i>5. Further Discussion</i>	<i>28</i>
<i>6. Conclusion and Implications</i>	<i>30</i>
<i>References</i>	<i>31</i>
<i>Appendix – Ethics Form.....</i>	<i>Error! Bookmark not defined.</i>

Table of Figures

Figure 1 : Distribution of DEA Efficiency Scores (VRS)	17
Figure 2 : Least Performing Trust by Efficiency Score	18
Figure 3 : Top Trusts Ranked by MCDA Scores	22
Figure 4 : DEA vs MCDA Sensitivity Analysis	23
Figure 5 : Number of Trusts by Efficiency Category	25

Table of Tables

Table 1 : Inputs and Output Variables	13
Table 2 : Best Performing NHS Trusts	16
Table 3 : Least Performing NHS Trusts	16
Table 4 : Profit/Loss Margins of Top Performing Trusts	18
Table 5 : Profit/Loss Margins of Least Performing Trusts	19
Table 6 : Performance of Trusts by MCDA Scores	21

Enhancing Healthcare Resource Allocation in England with Data Envelopment Analysis and Behavioral Principles

Abstract

The effective and fair distribution of healthcare resources continues to be a significant challenge for health systems globally, particularly within England's National Health Service (NHS). To address these challenges, this thesis presents a mixed-methods framework that combines Data Envelopment Analysis (DEA) for quantitative assessment of efficiency with behavioural principles to reflect human and organizational dynamics in the allocation of resources. This research evaluates hospitals based on financial and operational metrics; it also assesses the influence of cognitive biases and organizational culture on analytical results. The study demonstrates that integrating DEA with behavioural intelligence results in models that are analytically sound, contextually relevant, and practical. By connecting technical evaluation and practical decision-making, the results offer direction for policy and management, promoting transparent, flexible, and equitable distribution of NHS resources.

Keywords : *NHS, Healthcare resource allocation, Data Envelopment Analysis, , Behavioural sciences, Mixed methods, Organizational decision-making.*

1. Introduction

As healthcare systems strive to improve quality, efficiency, and equity in service delivery, the efficient allocation of healthcare resources is a paramount challenge. Increasing demands with squeezed budgets make it ever more important for decision-makers to embrace approaches that tackle effectiveness in a manner truly reflective of the dynamics of organizational behaviour. This thesis introduces a combined analytical method that combines Data Envelopment Analysis (DEA) with essential principles from behavioural sciences, intending to generate practical and contextually relevant insights for the allocation of healthcare resources.

1.1 Background on the NHS

Efficient allocation of healthcare resources is the foundational pillar of any health system across the world, especially in settings of escalating demand, constrained finances, and organizational complexity. Founded in 1948, the largest public healthcare system in England, the National Health Service (NHS), serves millions every year, thereby forwarding the prime determinant of service quality and cost-effectiveness, especially in finance, infrastructure, and workforce.

The NHS is an internationally acknowledged and publicly funded healthcare system that is known for providing health services, managing numerous trusts and vast networks in primary, secondary, and community healthcare environments. However, the distribution of resources within the NHS is plagued

by ongoing difficulties. These concerns are among the many issues regarding resource allocation under the NHS. The policy and academic discourse provide a more realistic depiction of the various deep-seated conflicts over supply and demand, as well as efficiency and equity in health provision (Charlesworth et al., 2022; Dixon et al., 2007). Certain resource distribution formulae have sometimes been criticized for being too unclear and further propagating inequity across regions when deprivation and unmet need are considered (Cookson et al., 2016). Longstanding issues include chronic underfunding in comparison to other European systems, chronic workforce shortages, increased costs, an ageing population, and widening regional funding gaps, making the limited NHS resources even more scrutinized (Dixon et al., 2003).

1.2 Current Analytical Approaches

To address the complexities of healthcare resource allocation, health economists have developed rigorous quantitative frameworks, with Data Envelopment Analysis (DEA) being one of the most prominent methods. However, before examining DEA's applications and advantages, it is essential to understand the broader landscape of quantitative methodologies previously utilised for healthcare resource allocation analysis.

Stochastic Frontier Analysis (SFA) is one parametric alternative that has been extensively compared with DEA in healthcare efficiency studies. In contrast to the DEA approach, SFA requires the specification of a functional form for the efficiency frontier and makes assumptions about the distributions of random error and inefficiency error terms (Katharakis and Katostaras, 2012). While SFA can account for statistical noise and measurement error, which DEA cannot handle directly, this parametric requirement can be restrictive in healthcare settings where the true production function is complex and unknown (Newhouse, 1994). Empirical comparisons show that DEA and SFA often yield divergent efficiency estimates, with SFA typically producing different rankings of healthcare units due to its underlying distributional assumptions.

Direct and indirect utility measurement techniques provide yet another approach to resource allocation, particularly in health technology assessment contexts. Findings from these methods emphasize measuring health state utilities with the use of standard gamble, time-trade-off, and visual analogue scales, or by means of overall quality of life questionnaires like EQ-5D and SF-6D. These methods are helpful in determining cost-effectiveness but are mostly limited to specific decision-making situations, lacking the broad organizational efficiency measures healthcare managers require for operational decisions.

1.3 Limitations of Existing Methods

Scholars do point out the inherent complexity in healthcare resource allocation and, indeed, warn that quantitative models are not sufficient unto themselves to grasp the sum total of multifaceted organisational decision-making realities (Hollingsworth, 2008). In spite of the big advances in

quantitative methodologies of Data Envelopment Analysis (DEA), existing approaches mostly discount behavioural factors, namely cognitive biases, organisational culture, and decision-making heuristics, which exert a serious impact on resource allocation in practice (Emrouznejad and Yang, 2018).

Numerous studies have shown that while DEA and similar quantitative methodologies furnish useful objective assessments, they often neglect to capture the less tangible behavioural ones (Kunc, 2018). This weakness results in technical approaches that are often robust but not necessarily complete when addressing reality in terms of decision-making environments (Brailsford et al., 2013). Current literature reviews have further emphasized that operational research in healthcare is often predicated on an assumption of rational behaviour, overlooking behavioural dynamics present among both staff and decision-makers (Djulgovic and Guyatt, 2017).

1.4 Model Selection

Such complex decisions have been aided by health economists in the development of rigorous quantitative frameworks, among which prominent is **Data Envelopment Analysis (DEA)**. This method is widely appreciated for assessing the efficiencies of various healthcare providers, including hospitals and NHS trusts, on multiple input-output conditions without determining any specific functional form (Mirzazaffari, Akbarzadeh & Ebrahimi, 2023; Jacobs, 2001; Hao et al., 2023). This nonparametric form has become valuable for benchmarking, best practice identification, and developing empirical evidence for resource allocation within the NHS. Nevertheless, scholars also point out the inherent complexity in healthcare resource allocation and, indeed, warn that quantitative models are not sufficient unto themselves to grasp the sum total of multifaceted organisational decision-making realities (Mirzazaffari, Akbarzadeh & Ebrahimi, 2023; Hao et al., 2023).

Despite these alternatives, DEA emerges as particularly well-suited for NHS resource allocation analysis due to several distinctive characteristics. DEA's nonparametric approach offers significant advantages in healthcare settings where the relationship between inputs and outputs is complex and often unclear, eliminating the need for restrictive functional form assumptions required by SFA (Cylus et al., 2017). The methodology uses linear programming techniques to construct an efficiency frontier based on the best-performing units in the dataset, allowing each organization to be evaluated against this empirically-derived benchmark rather than theoretical assumptions.

The ability to handle several inputs and outputs at the same time is another important advantage that sets DEA apart from other methods. This ability is crucial in healthcare, where hospitals use multiple resources like medical personnel, beds, equipment, and financial resources to generate different results, such as patient care, recovery rates, and quality metrics (Chakraborty et al., 2023). Unlike MCDA, which requires complex weighting procedures, or utility-based methods that focus on single outcomes, DEA naturally handles this multi-dimensional complexity through its mathematical programming approach.

1.5 Overview

This thesis is inspired by a significant need for strategies that surpass merely quantitative or policy-oriented frameworks. This research fundamentally focuses on how the principles of behavioural science, like cognitive biases, decision heuristics, and the influence of organizational culture, are related and shape resource allocation decisions within the NHS. In reality, decision-making is often swayed by behavioural constraints, straying from logical, model-driven suggestions. Opposition to change, bias towards the current situation, and poorly aligned incentives are common traits that can weaken or distort allocation strategies.

A key argument of this study is that integrating behavioural principles with quantitative efficiency analysis offers a more comprehensive and actionable framework for optimising healthcare resources. By explicitly addressing the human and organisational factors that influence resource allocation, it becomes possible to design strategies that are not only more effective, including one in terms of equity and efficiency, but also one that is more sustainable and implementable in real-world NHS settings.

The subject of this thesis is the complementary application of Data Envelopment Analysis and behavioural principles to improve the allocation of healthcare resources within an NHS context. By combining the quantitative measurement of efficiency with an appreciation of behavioural influences on the decision-making process, the present research will attempt to derive practicable yet analytically sound recommendations. The work will include a review of the literature, conduct empirical analyses of NHS trust data, and consider the implications of the integration of these approaches for policymaking and practice. This work aims to contribute to strengthening both the theoretical developments and the bridge linking the analytical models to healthcare management practice. An integrated summary with implications for the identified NHS systems will be provided and future research and implementation directions will be suggested in the last chapter.

2. Literature Review

Despite significant advances in quantitative methodologies like Data Envelopment Analysis (DEA) for assessing healthcare efficiency, existing approaches often discount behavioural factors—such as cognitive biases, organisational culture, and decision-making heuristics—that seriously impact resource allocation in practice. A number of studies have shown that while DEA and similar quantitative methodologies furnish useful objective assessments, they often neglect to capture the less tangible behavioural ones. This weakness results in technical approaches that are often robust but not necessarily complete when addressing reality in terms of decision-making environments (Hollingsworth, 2008; Emrouznejad & Yang, 2018; Kunc, 2018).

Current literature reviews have further emphasized that operational research in healthcare is often predicated on an assumption of rational behaviour, overlook behavioural dynamics present among both staff and decision-makers, although mounting evidence suggests that such factors are the real drivers behind resource allocation and policy implementation in the real world (Kunc, 2018).

There are also calls for integrating quantitative and behavioural evidence in healthcare research, with scholars and policy bodies recommending mixed methods approaches for the generation of holistic, transparent, and actionable insights for healthcare organizations (Noyes et al. 2019; Guetterman et al. 2015). This continual disconnection emphasizes the critical need for research that systematically incorporates sound quantitative assessment within the framework of behavioural science to bridge the divide between objective measurement and multifaceted decision-making in practice.

- **How can blended approaches that combine quantitative analysis with behavioural sciences be developed and applied to produce more holistic, transparent, and actionable resource allocation models in healthcare organisations?**

As a non-parametric method, Data Envelopment Analysis (DEA) serves as the prime determination of efficiency for healthcare organizations including NHS hospitals. It was developed by Charnes et al. (1978). DEA compares the efficiency of decision-making units utilizing a set of inputs against outputs without the need for a predetermined functional form. Early work by Nunamaker (1983) and Sherman (1984) enabled broad use of DEA in evaluating hospital performance. In the NHS, DEA is often combined with other methods, such as Stochastic Frontier Analysis, to benchmark hospital and primary care efficiency (Jacobs et al., 2006). Reviews highlight DEA's flexibility and its suitability for public sector analysis (Hollingsworth, 2008), while recent studies have refined models to better address quality, organisational factors, and statistical adjustments (Ozcan, 2014). Nevertheless, challenges exist with DEA, including its sensitivity to variable selection, outliers, and measurement error, limited guidance on model setting, and integrating qualitative factors like patient experience (Emrouznejad et al., 2018; Hollingsworth, 2008; Smith et al., 2005).

- **How can Data Envelopment Analysis be used to analyse and improve the efficiency of NHS hospitals while addressing model sensitivity and incorporating qualitative factors?**

The field of behavioural science has seen a surge in interest within health policy and practice, offering frameworks such as the Health Belief Model, Theory of Planned Behaviour, and behavioural economics to enhance the understanding of decision-making (Ajzen, 1991; Kahneman, 2011). However, effectively implementing these insights is essential for achieving lasting organizational change. The actions of healthcare professionals and patients are profoundly influenced by cognitive biases and environmental cues, which in turn shape resource allocation and compliance with established guidelines (Tversky & Kahneman, 1974). Although strategies such as customized feedback, reminders, and prompts have

shown promise in enhancing efficiency (Thaler & Sunstein, 2008), their effectiveness frequently relies on the specific context and may encounter constraints regarding scalability and sustainability. Within the NHS, tools like default choices and peer comparison letters demonstrate potential but have not seen broad implementation, highlighting obstacles like institutional resistance and conflicting priorities (Hallsworth et al., 2016). Global guidelines emphasize that purely rational methods are inadequate, promoting the systematic and context-sensitive incorporation of behavioural principles to tackle the intricate cognitive and cultural elements that affect decision-making (World Health Organization, 2021). Therefore, behavioural science provides important tools, yet their effective application demands careful implementation suited to the specific circumstances of healthcare organizations.

- **How can behavioural science interventions (such as nudges and feedback) be contextually integrated into the NHS to overcome cognitive and institutional barriers, thereby achieving sustainable improvements in decision-making and resource allocation?**

Efficient resource allocation is essential for enhancing patient outcomes and optimizing healthcare delivery; however, the integration of behavioural and stakeholder factors remains a challenge for conventional allocation techniques. Multi-Criteria Decision Analysis (MCDA) has become a leading approach to aid in complex decision-making scenarios, particularly when different and occasionally opposing goals must be harmonized in a clear and methodical manner (Angelis et al., 2016).

Angelis, Kanavos & Drummond (2016) emphasize that MCDA's flexibility allows both quantitative and qualitative criteria to be incorporated, ranging from measurable outcomes to preferences, values, and behavioural drivers. They note that MCDA frameworks make trade-offs explicit among conflicting objectives and help in structuring the uncertainty usually embedded in both criteria selection and weighting. The most popularly utilized MCDA models in healthcare improved transparency in decision-making. This is evident in Gongora-Salazar et al. (2023), who demonstrated through a systematic review that there are several limitations, including inconsistencies in methodology in association with the behavioural aspects not standardized in the capturing.

- **How can Multi-Criteria Decision Analysis (MCDA) be effectively utilized to combine both quantitative and behavioural criteria to address methodological inconsistencies in healthcare resource allocation decisions?**

The literature highlights the limitations of quantitative resource allocation methods in capturing the complexity and non-linearity of healthcare decision-making systems. Data Envelopment Analysis (DEA) furnishes an objective benchmark of efficiency but rarely considers behavioural and organisational dynamics that are often critical in influencing any resource use in practice – hence cognitive biases and cultural norms (Hollingsworth, 2008; Emrouznejad and Yang, 2018). Although Multi-Criteria Decision Analysis (MCDA) can include both quantitative and qualitative criteria, many

studies still speak of these problems on transparency, subjective weighting, and the appropriate incorporation of behavioural variables (Oliveira et al. 2019; Angelis et al., 2020).

The usefulness of behavioural science models in guiding healthcare improvement initiatives has grown over time, especially when it comes to addressing the organisational, cognitive, and motivational aspects that support practical decision-making in a healthcare environment. **Previously, Behavioural principles namely the Theory of Planned Behaviour (TPB), the Behaviour Change Wheel (BCW), and nudge-based interventions (Thaler & Sunstein, 2008; Horne et al., 2021) provide structured methodologies to design, implement, and evaluate interventions that go beyond purely rational, model-driven assumptions.** Research evidence that such models can be effectively employed to promote sustainable behavioural change in NHS contexts, for example by leveraging structured feedback, persuasive communication, and digital prompts to enhance operational adherence and improve organisational outcomes (Hurley et al., 2024). Quantitative efficiency analyses can therefore be translated into context-sensitive, useful strategies that take into consideration the realities of individual psychology and workplace atmosphere by incorporating behavioural science models into concrete recommendations.

The strongest of these responses have been nudges and feedback constructions, which are changing some staff behaviours but often get conditioned in the NHS by institutional and system-wide challenges (Hallsworth et al., 2016). Hence, the literature advocates the development of transparent blended methodologies that robustly imbed behavioural principles with quantitative models in the effort to implement better actionable resource allocation in healthcare settings.

In summary, the research questions for this study are as follows :

- **How can blended approaches that combine quantitative analysis with behavioural sciences be developed and applied to produce more holistic, transparent, and actionable resource allocation models in healthcare organisations?**
- **How can Data Envelopment Analysis be used to analyse and improve the efficiency of NHS hospitals while addressing model sensitivity and incorporating qualitative factors?**
- **How can behavioural science interventions (such as nudges and feedback) be contextually integrated into the NHS to overcome cognitive and institutional barriers, thereby achieving sustainable improvements in decision-making and resource allocation?**
- **How can Multi-Criteria Decision Analysis (MCDA) be effectively utilized to combine both quantitative and behavioural criteria to address methodological inconsistencies in healthcare resource allocation decisions?**

3. Research Methodology :

3.1 Model Evaluation

Beginning with model evaluation examines the combination of **DEA** with behavioural analysis, i.e., **Multi-Criteria Decision Analysis (MCDA)**, helping us understand the model's behaviour, assumptions, and limitations before evaluating its results and performance indicators.

The theoretical foundation for combining **Data Envelopment Analysis (DEA)** with **Multi-Criteria Decision Analysis (MCDA)** is one that addresses the inherent limitations of single-method approaches in healthcare. **DEA** excels at identifying relative efficiency frontiers and benchmarking organizational performance against best-practice peers, yet it operates within a purely quantitative framework that may overlook qualitative factors critical to healthcare delivery (Charnes ET AL., 1978; Cooper et al., 2007). **MCDA**, conversely, provides structured approaches for incorporating multiple, often conflicting criteria, including qualitative dimensions such as patient satisfaction, clinical quality indicators, and strategic alignment with organizational objectives (Belton & Stewart, 2002; Figueira et al., 2005). These methodologies amalgamate into a complete framework for measuring healthcare decision-making environments afflicted by mathematical rigor. The rationale for the integration highlights an environment characterized as stakeholder-heavy, where organizations should trade efficiency gains against quality, accessibility, equity, and other strategic objectives that traditional DEA models often overlook.

Model sequencing, criteria weighting, and integration would have to be considered at the initial stages to ensure that the DEA-MCDA framework has theoretical coherence and real-life applicability in terms of methods and methodology designed for it. The proposed approach utilizes **DEA as the primary efficiency measurement tool**, generating quantitative efficiency scores that serve as one criterion within a broader MCDA evaluation matrix (Greco et al., 2005). The framework's strength lies in its ability to maintain DEA's peer-based benchmarking capabilities while extending the evaluation scope to include strategic, qualitative, and stakeholder-relevant criteria. **Criterion selection** for the MCDA component must be multi-dimensional in line with the performance of healthcare, including clinical quality data, experience scores for patients, measures for engagement, financial sustainability indicators, and how health is seen in a community. Therefore, weight allocation has to also reflect the organizational priorities and stakeholder preferences, as mentioned in the **NHS Confederation** website. This flexibility allows for this approach to fit different organizational contexts with differing policy priorities and strategic interests while maintaining analytical consistency.

The integration between DEA and MCDA components does involve both methodological opportunities and potential validation challenges that must be carefully addressed. The most straightforward integration approach treats DEA efficiency scores as quantitative inputs to the MCDA, where they

compete with other performance criteria in determining overall organizational rankings (Despotis, 2005; Li & Reeves, 1999). **This method maintains the mathematical integrity of both techniques while also providing an exhaustive performance assessment that considers efficiency alongside other strategic objectives.** These alternatives include using MCDA for weighting DEA's input-output, using MCDA for selection of peer groups for DEA comparison, or DEA results for MCDA criteria development. Each of these integration pathways would require some validation mechanisms to ensure the framework can produce meaningful, actionable results that would surpass those obtainable from the single methodologies. The validation process must demonstrate that the integrated model provides superior decision support compared to standalone DEA or MCDA applications, generates insights not available through individual methodologies, and maintains theoretical coherence between the component approaches (Stewart, 1996).

Anticipated model performance characteristics of the DEA-MCDA framework should reflect the complementary strengths of both methodological approaches while mitigating their individual limitations. The **DEA component is expected to provide precise efficiency measurements**, clear benchmarking targets, and specific improvement recommendations based on peer group analysis (Hollingsworth, 2008; O'Neill et al., 2008). **Efficiency scores should demonstrate reasonable differences across NHS Trusts**, with a balance between units achieving optimal efficiency and those requiring significant improvement. The model should successfully identify best-practice organizations that can serve as learning targets for lower-performing Trusts. **The MCDA component should demonstrate its value by incorporating performance dimensions** that DEA cannot adequately address, potentially revealing that certain technically efficient Trusts still perform poorly on quality or strategic alignment measures (Dolan et al., 2005). The integrated framework should produce overall performance rankings that differ meaningfully from pure efficiency rankings, indicating that the additional criteria provide valuable supplementary information.

The expected outcomes of this integrated evaluation should demonstrate clear improvements over existing NHS performance measurement approaches while providing actionable insights for healthcare improvement initiatives. The model should successfully identify high-performing Trusts across multiple performance dimensions, enabling targeted learning and best-practice sharing programs (Jacobs et al., 2006). Conversely, **it should pinpoint organizations requiring improvement interventions and provide specific guidance on priority areas for development.** The framework should be equipped to provide more refined decisions on the allocation of resources, balancing efficiency against quality, access, and strategic objectives. Policy makers can use the model results to design specific support programs, modify allocation formulas, and assess the overall effectiveness of system improvement initiatives (Smith & Street, 2005). It is expected that an integrated approach will be superior to purely technical-efficiency measures in stakeholder acceptance, since it includes other dimensions of

performance which healthcare professionals, patients, and communities value. In this context, **a DEA-MCDA approach should contribute to enhancing NHS system performance by facilitating a performance assessment that is detailed**, rigorous, and informative in an evidence-based approach, recognizing that healthcare value creation in itself is overly complex and multidimensional.

3.2 Data Source

In this study, the data collection process centred on compiling a comprehensive dataset of NHS trusts, focusing on both financial and operational variables linked to capital investments, cost savings, and income generation activities. The data were sourced from the ‘ERIC’ excel file on the NHS Digital website that was published for the year 2023-2024.

Quantitative inputs for the analysis were captured through the following variables:

- **Capital investment for new build (£) - (cap_new_build)**
- **Capital investment for changing/improving existing buildings (£) - (cap_improve)**
- **Capital investment for maintaining (lifecycle) existing buildings (£) - (cap_maintain)**
- **Capital investment for equipment (£) - (cap_equipment)**
- **Energy efficient schemes costs (£) - (energy_costs)**

Capital investment for new build, improvement, maintenance, and equipment, as well as energy efficient schemes costs, were selected because they represent the core resource commitments NHS trusts make to sustain, expand, and modernize their physical infrastructure and equipment. These categories comprehensively cover the capital-expenditure areas of trusts within a strategic domain, which, in turn, influence the ability to provide, and the quality of, healthcare service provision. Energy-efficient investments address present needs while simultaneously targeting future sustainability goals.

Outputs, representing returns and efficiency outcomes of these investments, included:

- **Total contributions (£) (total_contrib)**
- **Waste re-use scheme - Cost savings (£) - (waste_savings)**
- **Income from car parking - patients and visitors (£) - (car_park_income)**

Total contributions, waste re-use scheme cost savings, and car park income were included as they collectively reflect the tangible financial returns and operational benefits derived from resource investments. Total contributions capture direct financial inflows linked to services provided; waste re-use savings indicate gains from improved resource stewardship and sustainability initiatives; and car park income offers a measure of how trusts generate ancillary revenue from essential service provision for patients and visitors. These outputs were chosen to directly align the evaluation of efficiency with

measurable financial outcomes and sustainability, ensuring comparability and relevance across trusts. While this thesis does focus on the recent events of the government reforms aimed at reducing NHS bureaucracy The dataset for the year 2024 to 2025 is yet to be published, only in October 2025, as mentioned on their website.

3.3 Data Processing

As mentioned previously in the literature review: limitations exist in DEA, which meant there had to be a proper preparation of the data to address the known limitations of Data Envelopment Analysis (DEA) in healthcare efficiency research.

First, variables included in the DEA models were carefully chosen based on their direct relevance to NHS operational practice. Inputs and outputs were limited to consistently defined, quantitatively robust measures found across all trusts, and their selection was justified with reference to both empirical research and NHS priorities. Sensitivity analyses such as varying the set of included variables and comparing efficiency scores were carried out to ensure the robustness of findings against arbitrary selection decisions.

To mitigate the influence of outliers and reduce measurement error, both of which can distort DEA results, numerous steps were taken to deal with the raw data. The raw data underwent pre-processing that included numeric type conversions, identification and handling of missing or anomalous values, and where necessary, transformation or exclusion of extreme datapoints. Descriptive statistics and visual inspection aided in detecting irregularities. For zero or missing values in essential input or output categories, a systematic approach was applied (such as substituting zeros for missing entries and other string values) to ensure the completeness of data, as recommended in best practice guidelines.

Table 1: Inputs and Output Variables

Variable Name	Description	Used in DEA	Used in MCDA
cap_new_build	Capital investment for new building projects (£)	Yes	No
cap_improve	Capital investment for changing/improving existing buildings (£)	Yes	No
cap_maintain	Capital investment for maintaining existing buildings (£)	Yes	No
cap_equipment	Capital investment for equipment (£)	Yes	No
energy_costs	Energy-efficient schemes costs (£)	Yes	No
total_contrib	Total financial contributions or income (£)	Yes	Yes
waste_savings	Waste re-use scheme cost savings (£)	Yes	Yes
car_park_income	Income from car parking (£)	Yes	Yes
efficiency_vrs	DEA VRS efficiency score	Output	Yes
efficiency_crs	DEA CRS efficiency score	Output	Optional
scale_efficiency	Scale efficiency (CRS / VRS)	Output	Yes
total_revenue	Aggregate revenue (sum of outputs)	Output	Yes

total_investment	Aggregate investment (sum of inputs)	Output	Yes
net_profit_loss	Net profit/loss (revenue minus investment)	Output	Yes

The inputs and outputs variables as shown in Table 1 provide a clear overview of the key data points used in the analysis to evaluate NHS Trust performance. **Inputs consist of several types of capital investments and operational costs, such as spending on new buildings, improvements, maintenance, equipment, and energy efficiency measures.** These represent the resources or costs incurred by each Trust. **Outputs, on the other hand, capture the financial and operational results generated by those inputs, including total monetary contributions, savings from waste reuse programs, and income from car parking services.** This distinction between inputs and outputs allows the efficiency analysis to measure how well each Trust converts resources into valuable financial and operational outcomes, forming the basis for the DEA calculations and the subsequent multi-criteria decision analysis (MCDA).

4. Analysis & Discussions

The analysis is led by the application of **Data Envelopment Analysis (DEA)** which is a quantitative tool for the evaluation of the relative efficiency of multiple decision-making units (DMUs) in this case: NHS Trusts. **DEA is a powerful mathematical analysis to assess the relative efficiency of a set of DMUs, such as hospitals, universities, and companies, which operate in a multi-input and multi-output environment. This analysis is based essentially on the ratio of weighted outputs to weighted inputs for each DMU and compares them with the most efficient DMUs in that group.**

DEA does not require predetermined weights, allowing the data to reveal which units are performing optimally. DEA establishes which DMUs are efficient by devising an efficiency frontier. In addition, it finds out the scale or resource combinations that will allow other DMUs to potentially improve their performance (Charnes, Cooper, & Rhodes, 1978; Shero et al., 2021).

Within DEA, two common models for returns to scale are Variable Returns to Scale (VRS) and Constant Returns to Scale (CRS):

- **Constant Returns to Scale (CRS):** The assumption means that increasing all inputs by a certain proportion (for example, doubling all resources) will lead to an increase in outputs in exactly that proportion (wherein outputs also double). CRS is suitable when we think that organizations operate at an optimal size: that is, they are equally efficient in scaling up or down their operations.
- **Variable Returns to Scale (VRS):** The presumptions of the input-output relationship are evaluated here. Under VRS, increasing inputs may cause output to increase by greater amounts, by the same amount, or by lesser amounts. In other words, as the scale of the operation changes,

efficiency may either improve (increasing returns), remain the same, or decline (decreasing returns). VRS is useful for organizations of varying sizes or when not all organizations are believed to be operating at their most productive scale.

The **Constant Returns to Scale (CRS)** assumption states that if an NHS Trust were to increase all its resource inputs, such as capital for new buildings, improvements, maintenance, equipment, and energy efficiency, by a certain proportion, it would experience a precisely proportional increase in its outputs. These outputs in the dataset include financial contributions, waste re-use savings, and car park income. CRS therefore applies best when it is considered that all the NHS Trusts are assumed to be at an optimal operational size, i.e., whether they are on a larger or smaller scale, it does not matter, for no inherent advantage or otherwise shall become available through scaling. In this situation, the efficiency score of any Trust refers to its ability to turn resources into good outcomes and is not affected by the confounding of scale. By this criterion, all Trusts are placed on a common platform of comparability across the board despite differences in sizes.

The **VRS model**, on the other hand, allows for the possibility that the efficiency of an NHS Trust may change with a change in its operational scale. Under VRS, an increase in all inputs may give an increasing return with an output that increases by more than the proportional increase of inputs, constant return with an output that increases exactly in proportion to the increase in input level, or decreasing return, with an output level that increases less than the proportional increase of inputs. This flexibility makes the VRS approach particularly adept for analysing the database as not all Trusts are necessarily functioning on the optimal scale. Some of them might benefit from economies of scale while others might have experienced inefficiencies due to managerial or structural issues as their operations expand or contract. The VRS model tells a story of efficiency that relates to nuances in size and context; it makes it much easier to determine where NHS Trusts stand regarding both technical inefficiency and suboptimal scale.

4.1 Model Results

4.1.1 Measuring Efficiency and Benchmarking

DEA forms the backbone of the initial assessment, relying on both the **Variable Returns to Scale (VRS)** and **Constant Returns to Scale (CRS)** models to benchmark NHS Trusts. Each Trust is compared not to an external standard but to a peer-defined efficiency frontier, set by the best-performing Trusts. The VRS model, which considers technical inefficiency and scale inefficiency separately, yields a spectrum of efficiency scores.

Top performers, such as “**Essex Partnership University**”, were calculated to have a score of 1, signifying best-practice status among peers. The scale efficiency, calculated as the ratio of CRS to VRS

efficiency, further disentangles inefficiency due to size from that intrinsic to operations, crucial insight for trusts wrestling with resource allocation amidst varying operational scales.

Table 2: Best Performing NHS Trusts

Trust Code	Trust Name	Efficiency VRS	Efficiency CRS	Scale Efficiency
RIL	Essex Partnership University NHS Foundation Trust	1	0.179	0.179
RGM	Royal Papworth Hospital NHS Foundation Trust	1	1	1
RAN	Royal National Orthopaedic Hospital NHS Trust	1	1	1
RNK	Tavistock And Portman NHS Foundation Trust	1	1	1
RWK	East London NHS Foundation Trust	1	0.066	0.066
RL1	Robert Jones And Agnes Hunt Orthopaedic Hospital NHS Foundation Trust	1	1	1
RLQ	Wye Valley NHS Trust	1	1	1
RT5	Leicestershire Partnership NHS Trust	1	0.042	0.042
RXM	Derbyshire Healthcare NHS Foundation Trust	1	1	1
RXT	Birmingham And Solihull Mental Health NHS Foundation Trust	1	1	1

The highest 10 performing NHS Trusts, as shown in the efficiency scores table above, all reached the maximum possible VRS efficiency score of 1. This classifies them as benchmark performers on the efficiency frontier, indicating they are maximizing their capital investments and operational resources like expenditures on new constructions, enhancements, upkeep, equipment, and energy-efficiency initiatives to generate optimal output in financial contributions, waste reductions, and car parking revenue compared to their counterparts. Trusts such as **ESSEX PARTNERSHIP UNIVERSITY**, **ROYAL PAPWORTH HOSPITAL**, **ROYAL NATIONAL ORTHOPAEDIC HOSPITAL**, among others in this top performing group, act as models; their procedures and distribution of resources are viewed as ideal, establishing benchmarks for what can be accomplished with existing evaluation metrics. Notably, some of these top trusts also reached a CRS efficiency score of 1, indicating both technical and scale efficiency, while a few showed lower scale efficiency, suggesting that although they manage their operations very well, there may still be some benefits to adjusting operational scale.

Table 3: Least Performing NHS Trusts

Trust Code	Trust Name	Efficiency VRS	Efficiency CRS	Scale Efficiency
R0D	University Hospitals Dorset NHS Foundation Trust	0.012	0.007	0.617
RC9	Bedfordshire Hospitals NHS Foundation Trust	0.011	0.008	0.699
RRK	University Hospitals Birmingham NHS Foundation Trust	0.011	0.007	0.664
RXK	Sandwell And West Birmingham Hospitals NHS Trust	0.011	0.011	0.951
RR8	Leeds Teaching Hospitals NHS Trust	0.009	0.003	0.279
RHM	University Hospital Southampton NHS Foundation Trust	0.009	0.007	0.777
RWE	University Hospitals of Leicester NHS Trust	0.009	0.006	0.703
RTH	Oxford University Hospitals NHS Foundation Trust	0.006	0.004	0.608
RX1	Nottingham University Hospitals NHS Trust	0.004	0.003	0.743

Conversely, the **table above displays the 10 worst performing trusts**, which exhibit exceptionally low **VRS efficiency scores**, typically varying from around **0.012 to as low as 0.004** for “**Nottingham University Hospitals**”. These low scores demonstrate considerable inefficiency, suggesting that these trusts are not effectively converting their resource inputs into equivalent outputs relative to similar organizations. The data indicates that these lowest achievers might reach similar output levels with significantly fewer resources, revealing considerable room for enhancement. Certain trusts in this group, including “**University Hospitals Dorset**” and “**Bedfordshire Hospitals**”, exhibit low scale efficiency, indicating both operational and structural shortcomings. These trusts represent promising targets for focused interventions and resource redistribution, as their inefficiency indicates missed chances for improved financial results and enhanced performance within the NHS framework.

4.1.2 Visualizing Performance and Identifying Extremes

The bar chart below showing the allocation of DEA efficiency scores demonstrates how NHS Trusts compare to one another regarding operational efficiency, as determined by the Data Envelopment Analysis (DEA) approach. In the chart, the horizontal axis (x-axis) displays the spectrum of efficiency scores from 0 (least efficient) to 1 (most efficient). The vertical axis (y-axis) indicates the count of Trusts within each efficiency score range, with each bar representing a particular interval (or "bin") of efficiency scores.

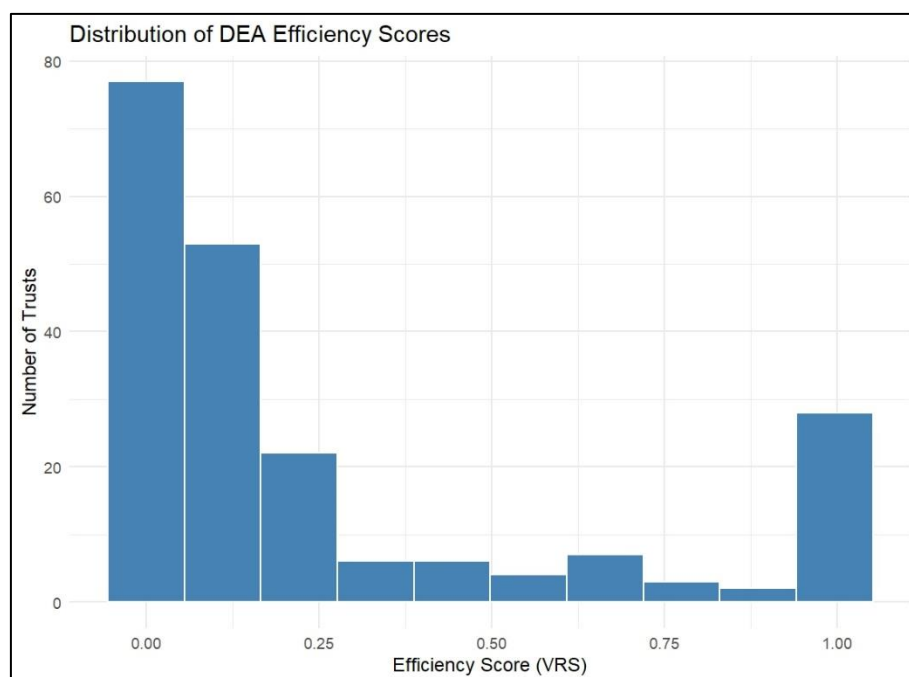


Figure 1: Distribution of DEA Efficiency Scores (VRS)

The visual reveals that the vast **majority of Trusts have low efficiency scores**, as indicated by the highest bars clustered near the left end of the x-axis. This indicates that many Trusts are comparatively less efficient than their top-performing counterparts. **A consistent decrease in Trusts occurs as the efficiency score rises, indicating that fewer organizations achieve or come close to higher**

efficiency levels. Notably, a smaller group of Trusts achieves a perfect score of 1, indicated by a prominent bar at the far right. These Trusts are considered operationally optimal and hence they set to be the "efficiency frontier" and are used as comparative benchmarks.

In conclusion, the distribution is **significantly skewed**: the majority of Trusts function with considerable inefficiency, whereas only a small number achieve complete efficiency. **This trend emphasizes considerable opportunities for enhancing performance among the underperforming majority and pinpoints a few top-performing organizations within the NHS framework.** The chart further supports the emphasis on specific interventions and the redistribution of resources for those Trusts situated in the lower efficiency categories.

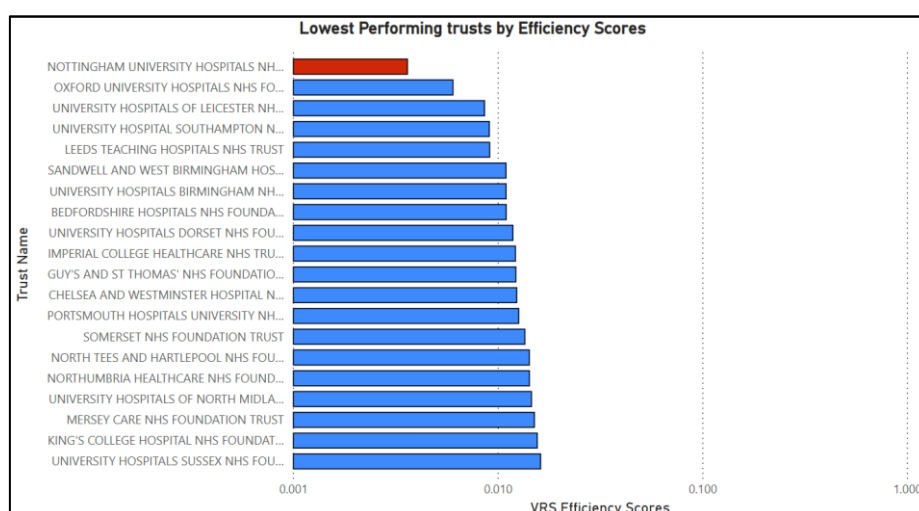


Figure 2: Least Performing Trust by Efficiency Score

For instance, it can be noted that “**Nottingham University Hospital**” appears as the most inefficient in Figure 2, merging significant investments with limited results. This diagnostic ability facilitates precisely focused performance enhancement efforts, as it identifies not only which Trusts are lacking but also the reasons (e.g., disparity between investment and returns).

4.1.3 Connecting Financial Sustainability: Profit/Loss Analysis

The profit and loss component adds a vital financial lens. Each Trust's net revenue (the sum of all positive and negative output values) is compared to its total investment, revealing the actual fiscal health alongside operational efficiency. This dual perspective is crucial: a Trust can be relatively efficient but still operate at a significant loss if the external environment or uncontrollable factors are at play.

Table 4: Profit/Loss Margins of Top Performing Trusts

Trust Code	Trust Name (NHS Foundation Trusts)	Scale Efficiency	Total Revenue	Total Investment	Net Profit/Loss
RX1	NOTTINGHAM UNIVERSITY HOSPITALS	0.7426	7,692,730	160,719,081	-153,026,351
RTH	OXFORD UNIVERSITY HOSPITAL	0.6081	11,080,198	80,038,000	-68,957,802

RWE	UNIVERSITY HOSPITALS OF LEICESTER	0.7027	10,161,594	59,753,156	-49,591,562
RHM	UNIVERSITY HOSPITAL SOUTHAMPTON	0.7772	7,553,714	46,680,000	-39,126,286
RR8	LEEDS TEACHING HOSPITALS NHS TRUST	0.2794	17,232,075	52,461,745	-35,229,670
RXK	SANDWELL AND WEST BIRMINGHAM HOSPITALS	0.9506	1,037,661	66,543,968	-65,506,307
RRK	UNIVERSITY HOSPITALS BIRMINGHAM	0.6642	9,955,199	45,520,824	-35,565,625
RC9	BEDFORDSHIRE HOSPITALS	0.6993	8,499,923	98,400,316	-89,900,393
R0D	UNIVERSITY HOSPITALS DORSET	0.6168	9,254,656	89,147,699	-79,893,043
RYJ	IMPERIAL COLLEGE HEALTHCARE	0.6745	7,511,646	76,745,699	-69,234,053

The bottom 10 NHS trusts in the profit/loss analysis table are those exhibiting the largest net financial deficits while also ranking among the least efficient according to Data Envelopment Analysis. At the forefront of this group is “**Nottingham University Hospital**” (RX1), which has reported a remarkable net loss of **-£153 million**, along with the lowest efficiency score in the dataset. Likewise, “**Bedfordshire Hospital**” (RC9) reports a **net loss of £89.9 million**, whereas “**Barts Health**” (R1H), “**University Hospitals Dorset**” (R0D), and “**London Northwest University Healthcare**” (R1K) each confront deficits between **-£73 million and -£83 million**. The trusts mentioned in the table above exhibit considerable scale inefficiencies, suggesting difficulties in effectively transforming resources into outputs. This ongoing, entrenched inefficiency, combined with substantial net losses and considerable yearly investments, reveals systemic issues that extend well beyond operational margins or occasional deficits. These findings indicate that in the absence of significant structural reforms, resource redistribution, and focused policy measures, these trusts will persist in experiencing unsustainable deficit cycles irrespective of their service size or complexity.

Table 5: Profit/Loss Margins of Least Performing Trusts

Trust Code	Trust Name (NHS Trust/ Foundation Trust)	Scale Efficiency	Total Revenue	Total Investment	Net Profit/Loss
RRU	LONDON AMBULANCE SERVICE	1	0	4,155,627	-4,155,627
RXQ	BUCKINGHAMSHIRE HEALTHCARE	0.0294	2,318,785	40,406,058	-38,087,273
RX9	EAST MIDLANDS AMBULANCE SERVICE	0.996	83,207	1,645,306	-1,562,099
RP6	MOORFIELDS EYE HOSPITAL	0.0723	674,495	45,217,789	-44,543,294
RAT	NORTHEAST LONDON	0.419	0	23,954,353	-23,954,353
RBD	DORSET COUNTY HOSPITAL	0.5713	1,110,111	18,406,199	-17,296,088
R1D	SHROPSHIRE COMMUNITY HEALTH	1	0	1,809,223	-1,809,223
RXY	KENT AND MEDWAY NHS AND SOCIAL CARE PARTNERSHIP TRUST	0.9026	2,052,771	9,860,000	-7,807,229
RMY	NORFOLK AND SUFFOLK	0.5446	560,499	8,752,207	-8,191,708
RRJ	ROYAL ORTHOPAEDIC HOSPITAL	0.9808	606,689	2,452,069	-1,845,380

The top 10 trusts listed in the profit/loss and efficiency table are those with the highest efficiency scores below the perfect value of 1, thereby avoiding inclusion of statistical outliers. The aforementioned trusts

namely “**London Ambulance Service**” (RRU), “**Buckinghamshire Healthcare**” (RXQ), and “**East Midlands Ambulance Service**” (RX9) each demonstrate remarkably high technical efficiency according to DEA. Despite this, all continue to post annual net losses, ranging from -£4.2M to as high as -£38M, indicating significant financial pressures even among highly efficient operations.

Many of these leading entities are ambulance or specialized trusts (e.g., “**Moorfields Eye Hospital**” and “**Dorset County Hospital**”), which typically face intricate service requirements yet handle their resources with remarkable efficiency. “**Royal Orthopaedic Hospital**” (RRJ) and “**Kent and Medway**” (RXY) also excel in maintaining a balance between moderate income and investments, achieving efficiency ratings significantly higher than the industry average.

Together, the trusts shown in table 5 demonstrate that effective management can exist alongside significant deficits, pointing to systemic funding issues rather than just operational inefficiencies, and emphasizing the necessity for ongoing investment and policy focus even for those achieving high operational performance.

4.1.4 Multi-Criteria Decision Analysis

Multi-Criteria Decision Analysis (MCDA) is executed in this section which basically consolidates the comparative performance of NHS Trusts by merging technical, financial, and operational scale viewpoints into one clear composite measure. The methodology and the formulas utilised to calculate the MCDA scores are as follows and will explain the results of the analysis:

Firstly, three core indicators were used for MCDA scoring:

- **Efficiency:** Normalized DEA efficiency score (Variable Returns to Scale, VRS)
- **Profit/Loss:** Normalized net profit/loss (reflecting financial health)
- **Scale Efficiency:** Normalized ratio of CRS to VRS DEA scores

Each indicator was normalized to a 0–1 range using the min-max normalization formula:

$$x_i^{norm} = \frac{x_i - \min(x)}{\max(x) - \min(x)}$$

Where :

- x_i is the raw value for the Trust “i”
- min/max span all the Trusts.

The final MCDA Weighted Score for each Trust was calculated using:

MCDA Weighted Score_i

$$= 0.5 \times Efficiency_i^{norm} + 0.3 \times \frac{Profit}{Loss_i^{norm}} + 0.2 \times ScaleEfficiency_i^{norm}$$

Where:

- 0.5 is the weight for normalized technical efficiency.
- 0.3 is the weight for normalized net profit/loss.
- 0.2 is the weight for normalized scale efficiency.
- The sum of all weights is 1.
- “norm” refers to the normalized value.

This weighting reflects the policy priority that technical efficiency is crucial, while also incentivizing good monetary management and scaling. After calculation, Trusts were ranked by their MCDA Weighted Score, making the results actionable for benchmarking and resource decisions. The top rankings were populated by Trusts excelling across all criteria demonstrating efficient services, strong economic management, and appropriate operational scale.

Table 6: Performance of Trusts by MCDA Scores

Trust Name (NHS FOUNDATION TRUST)	Efficiency	Profit/Loss	Scale Efficiency	MCDA Weighted Score
TORBAY AND SOUTH DEVON	1	0.957	1	0.987
QUEEN VICTORIA HOSPITAL	1	0.94	1	0.982
WIRRAL COMMUNITY HEALTH AND CARE	1	0.931	1	0.979
PENNINE CARE	1	0.922	1	0.976
WALTON CENTRE	1	0.913	1	0.974
WIRRAL UNIVERSITY TEACHING HOSPITAL	1	0.902	1	0.971
BRADFORD DISTRICT CARE	1	0.896	1	0.969
NORTHEAST AMBULANCE SERVICE	1	0.888	1	0.966
SOUTH TEES HOSPITALS	1	0.875	1	0.962
THE NEWCASTLE UPON TYNE HOSPITALS	1	0.867	1	0.96

For example, “**Torbay and South Devon**” achieved the **highest MCDA score (0.987)**, combining perfect technical efficiency and scale efficiency with an excellent profit/loss score. This sets an exemplary benchmark for peers.

Other leading Trusts, including “Queen Victoria Hospital” and “Wirral Community Health and Care”, exhibited a similar profile: high normalized scores across all three domains. The top 15 trusts with optimal MCDA scores are visualised in the form of a stacked bar chart as shown in Figure 3.

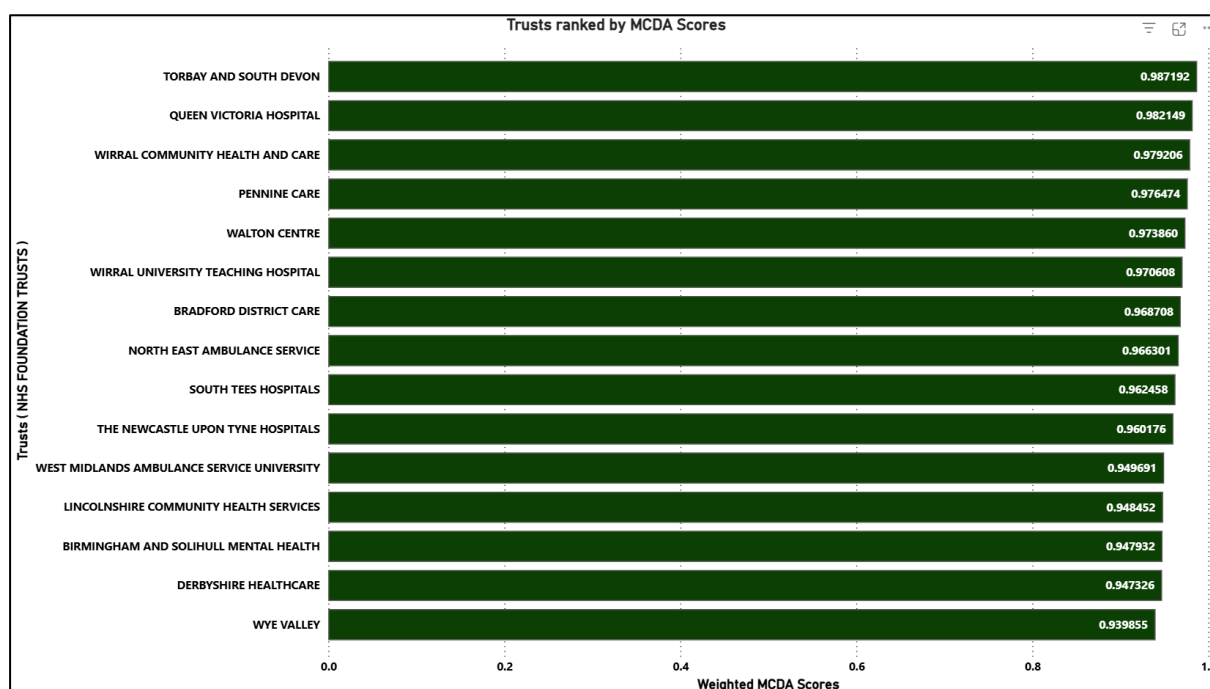


Figure 3: Top Trusts Ranked by MCDA Scores

4.1.5 Sensitivity and Scenario Analysis: Weighted DEA

Recognizing that not all inputs and outputs are equally strategic, the analysis includes a weighted DEA scenario. Strategic emphasis is given to variables like new build investments and total contributions by applying higher weights. Comparison of unweighted versus weighted results demonstrates the model’s stability which highlights that highly efficient Trusts remain so, but mid- and low-tier Trust rankings may shift, reflecting the organization’s alignment with specific strategic objectives.

Input Weights

- **Capital investment for new build (Weight: 2):**

New build capital is often the most significant and visible investment, leading to new or expanded facilities. By setting this weight to 2 (the highest among inputs), the analysis underscores the strategic emphasis on major infrastructure expansion aligning with long-term NHS service and estate transformation goals.

- **Capital investment for improvements/maintenance/equipment (Weight: 1 each):**

These are essential, but typically smaller in scale and strategic impact compared to new builds. They support quality, safety, and operational continuity, so each receives equal but lower weight

to recognize their importance while reflecting they are neither as strategic nor as transformational as full new-builds.

- Energy-efficient schemes costs (Weight: 1):

Energy savings and decarbonization are policy priorities, but their financial footprint is generally less than capital construction. Equal footing with maintenance and improvements ensures they are not neglected, while still weighting new build highest.

Output Weights

- Total contributions (Weight: 2):

This might include contributions, funding, and various external revenues. It holds the highest weight among outputs since capturing, utilizing, and enhancing external contributions is essential for financial sustainability and frequently a policy objective for NHS organizations.

- Waste savings and car park income (Weight: 1 each):

Both are real sources of savings and revenue, but their scale and strategic impact are typically less than total contributions. They support day-to-day efficiency and self-generated income, so they maintain a meaningful weight, but less than total inflows from broader contributions.

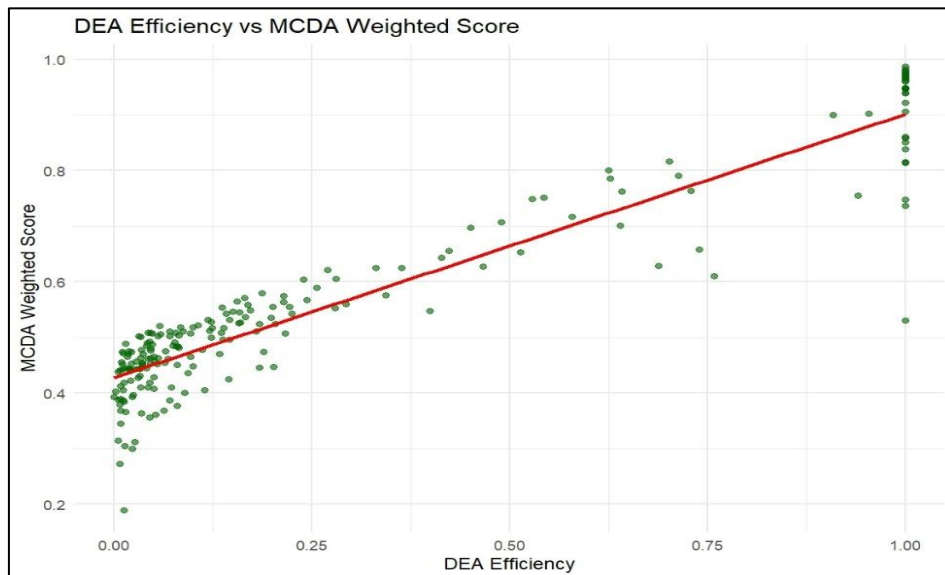


Figure 4: DEA vs MCDA Sensitivity Analysis

The plot displayed above illustrates the correlation between two key performance indicators computed for NHS Trusts namely DEA and MCDA . The efficiency scores of DEA for each trust is displayed on the x-axis and these indicate how efficiently a trust converts its input resources (such as capital investments and operational expenses) into outputs (like revenue, savings, and financial contributions), depending exclusively on relative performance compared to peers through nonparametric linear

programming. On the y-axis, the MCDA weighted score is plotted, and this extends the analysis by combining technical efficiency with other critical performance dimensions, such as financial health (profit/loss) and scale efficiency, using a tailored set of behavioural and operationally relevant weights.

Each **green dot represents an individual NHS Trust**, and the **red line is a fitted regression** indicating the overall relationship between the two scoring approaches. The plot clearly reveals a strong, positive correlation: **Trusts with higher DEA efficiency scores tend to have higher MCDA weighted scores as well**. Nonetheless, the variation around the regression line emphasizes that although numerous efficient **Trusts excel on the overall composite MCDA metric, there are cases where high technical efficiency (elevated DEA) does not ensure a similarly high MCDA score**. This happens because the MCDA scoring system considers extra elements such as ongoing profitability and ideal operational scale, which might not always align with the technical efficiency evaluated solely by DEA.

The visual pattern underscores the significance of utilizing both approaches together: **DEA offers a clear, input-output-centric perspective**, whereas **MCDA incorporates a broader array of performance and behavioural standards**. The clustering at high DEA and MCDA scores also indicates the presence of several optimally performing Trusts, whereas more dispersed points at lower DEA values demonstrate the diversity of organizational performance and the potential impact of other contextual or behavioural factors that MCDA captures.

4.2 Actionable Recommendations

The classification of NHS trusts according to levels of operational efficiency are categorised as 'Efficient', 'High', 'Medium', and 'Low' serves as an integral mechanism for the allocation of targeted organisational recommendations. While trusts occupying the 'Efficient' category are advised to maintain established practices, those falling within the 'Low efficiency' group, which notably comprise the majority (164 out of 205), are identified as requiring substantial reallocation of resources. The development of actionable recommendations should therefore be informed by empirically validated behavioural science principles, offering custom strategies to drive meaningful and sustainable improvements commensurate with each trust's efficiency standing.

In the context of **trusts within the bracket of high efficiency**, the **principle of positive reinforcement from Organisational Behaviour Management (OBM)** offers tangible benefit when implemented via structured recognition and reward systems. A relevant case would be the **'Royal Devon University Healthcare NHS Foundation Trust'**, where a comprehensive staff recognition strategy was implemented in response to **organizational restructuring**. This strategy included both formal methods, such as annual award ceremonies and recognition for long service, and informal methods that included thank-you schemes via intranet and frequent nominations by peers. Feedback from staff showed marked improvement in terms of morale and perceived organizational commitment, while engagement metrics

sustained improvements of workforce sentiment and job satisfaction. Such outcomes emphasize the importance of positive reinforcement in sustaining high-performance behaviours in healthcare organizations. (NHS Employers, 2025).

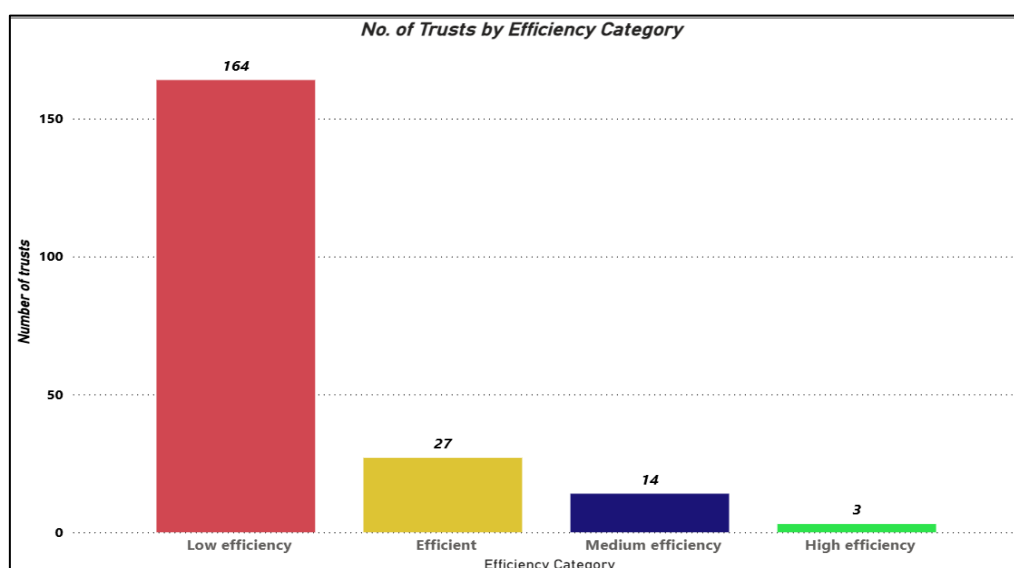


Figure 5 : Number of Trusts by Efficiency Category

Recent evidence underlines the efficacy of nudge-based interventions in health care settings, even as regards the outcomes of or improvements in adherence to medications or preventive health behaviours. One notable real-world example is the “**ENCOURAGE Randomized Controlled Trial**”, which utilised digital nudges such as personalised text messages and automated emails to improve adherence to statin therapy among patients with cardiovascular conditions. Participants received regular, encouraging prompts (e.g., congratulatory messages on successful medication routines and reminders for weekly adherence), resulting in a significant increase in medication compliance at twelve months compared to standard care (**66.3% adherence in the nudge group versus 50.5% in the control group**) (Horne et al., 2021).

For trusts categorised as “moderately efficient”, it is recommended to adopt a **nudge-based digital reminder system** to support staff and patient behaviours linked to efficiency objectives. For example, hospitals could use automated, supportive text messages sent through an internal communication platform about adherence to schedules, task completions, or recommended best practices in order to encourage continuous engagement in productivity-enhancing activity without restricting choice or increasing administration load. These subtle prompt messages could be configured to local specifications and evaluated to effect; a practically scalable intervention tied to behavioural science.

Behavioral models such as the **Theory of Planned Behaviour (TPB)** are proving ever more useful in contemporary health care management's efforts to design interventions targeting inefficiency in

organizations. **A concrete example of employing the TPB in practice within NHS trusts is in developing a human rights-based approach within acute inpatient mental health settings.** Empirical investigation in the Northwest of England demonstrated that mental health professionals' intentions to adopt the novel approach were principally shaped by their attitudes towards the intervention and by the perceived expectations of influential peers. Despite some organisational obstacles impeding perceived behavioural control, the provision of structured educational sessions to elucidate the benefits, the nomination of peer champions, and targeted skills training proved effective in enhancing staff intentions and readiness for change (Le Bruna et al., 2024). This evidence substantiates the value of leveraging all TPB constructs attitude, subjective norms, and perceived behavioural control when designing interventions aimed at improving organisational efficiency in healthcare contexts.

To effectively address low efficiency within trusts, the application of the **Behaviour Change Wheel (BCW)** provides a comprehensive and evidence-based framework. The **BCW-informed initiative** aimed recently at improving person-centred conversations concerning physical activity across clinical practice in NHS settings. **The intervention used principles of persuasion and education to provide healthcare professionals with the resources and skills required to improve their ability, opportunity, and motivation for routinely engaging in these conversations.** Barriers and enablers were systematically identified and addressed through a combination of tailored persuasive communication and structured skill-building, resulting in massive advances in the frequency and quality of such behaviour-change conversations, which are crucial for organizational improvement at a greater scale. (Hurley et al., 2022).

As for trusts that were less efficient, a multi-component intervention based on BCW would be a good recommendation. Such actions would ideally include targeted educational seminars to clarify the operational and clinical benefits of such efficient practice, advocacy using persuasive case examples to change professional attitudes, and empowering respected staff as champions. Strengthening such efforts can support them with solid support systems and further skill building workshops towards creating a more amenable environment for sustaining the behavioural change.

Across all efficiency categories, it remains vital to establish clear expectations and a continuous feedback infrastructure. By embedding regular communication about performance standards and by establishing continuous feedback loops, trusts reinforce positive behaviours and address inefficiencies directly. **When employees feel included and engaged, ownership and intrinsic motivation are created, which are also key tenets of comprehensive behavioural science approaches to organisational improvement.**

Ultimately it is to be noted that the efficacy of the various recommendations will depend on reaching a behavioural intervention that is already efficient for the trust and explicitly based upon rigorous behavioural principles; one for instance, nudging for gentle guidance, reinforcement for consolidation, interventions stemming from the working of TPB for foundational shifts ensuring a mix of immediate effect and sustained organizational transformation.

4.3 Summary and Impact

The analysis presented in previous sections revealed significant insights into the operational and financial efficiency of NHS Trusts using an integrative **Data Envelopment Analysis (DEA)** and **Multi-Criteria Decision Analysis (MCDA)** framework. The results, derived from carefully selected input and output variables, provide a robust basis for understanding how well NHS Trusts convert resources into financial and operational outcomes, while also highlighting systemic inefficiencies and actionable pathways for improvement.

The DEA results, as shown in Figures 1 through 4, quantify efficiency scores under both **Variable Returns to Scale (VRS)** and **Constant Returns to Scale (CRS)** assumptions. These scores measure how well each Trust utilizes inputs such as capital investments in new builds, maintenance, equipment, and energy-efficient schemes in order to generate outputs including total revenue, financial contributions, and waste savings. The outputs for each Trust were benchmarked against the best-performing peers, building an efficiency frontier that distinguishes exemplary Trusts from underperformers.

Highly skewed distribution was illustrated in the plots Figure 1 and the associated results table towards lower efficiency scores throughout the NHS Trusts, with the majority clustered at efficiency levels well below 1. This reflects a broad inefficiency problem across the system, underscoring the challenge for NHS leadership in closing operational and financial gaps. The top ten Trusts achieving perfect VRS efficiency scores of 1 (e.g., “**Royal Papworth Hospital**”, “**Royal National Orthopaedic Hospital**”) are characterized by optimal resource utilization and financial sustainability, setting performance standards by exemplifying effective management of capital and operational inputs.

The **MCDA integration** further contextualizes these findings by combining DEA scores with financial and operational metrics into a weighted composite score. Trusts such as “**Torbay and South Devon**” achieved the highest MCDA scores, reflecting balanced excellence across technical efficiency, scale efficiency, and profitability. This specific scoring method allows for more nuanced policy and management strategies while validating DEA results and considering broader organisational performance parameters.

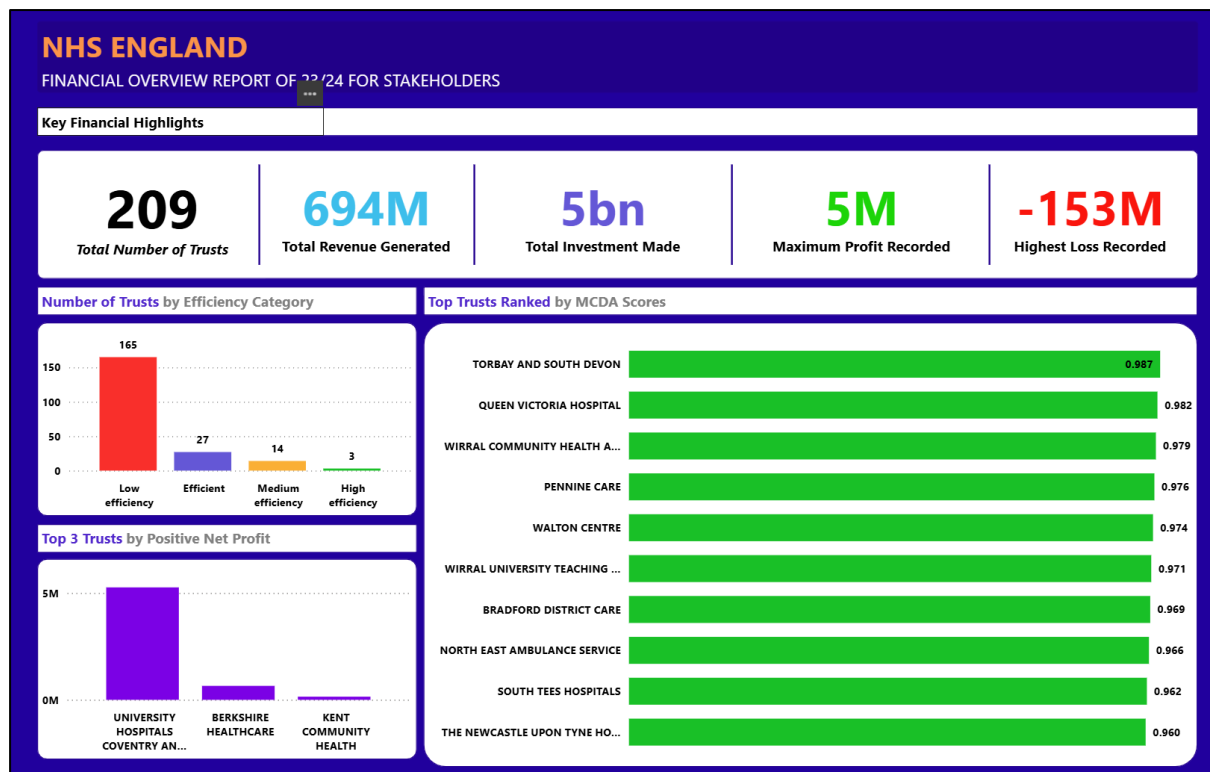


Figure 6 : Financial Dashboard of NHS England 23/24

A summary of key performance indicators (KPIs) was created in Power BI enveloped as a financial dashboard to highlight the essential facts. The total number of Trusts analysed was **209**, collectively generating a total revenue of approximately **£694.98 million**. However, the total investment made by these Trusts was significantly higher, upwards of **£5 billion**, reflecting substantial capital and operational resource commitments. Correspondingly, the **aggregated net profit/loss** for these Trusts resulted in a large deficit of approximately **-£686.03 million**, highlighting systemic financial challenges within the studied set. Among these, the highest loss recorded was by “**Nottingham University Hospitals**”, with a **net loss of -£153.03 million**, marking it as the most financially strained entity in this cohort. These KPIs underscore the scale of financial inefficiency and fiscal pressures facing several NHS Trusts, setting the context for targeted interventions and strategic resource reallocation to improve overall NHS sustainability and efficiency.

5. Further Discussion

This study has established a novel, integrated approach using **Data Envelopment Analysis (DEA)** and **Multi-Criteria Decision Analysis (MCDA)**, accompanied by behavioural principles, to benchmark and optimize NHS Trust resource allocation. While the study delivers valuable findings and actionable policy guidance, it is also marked by methodological limitations and points the way to several promising future research directions.

Further research could consider an updated dataset as the data is updated on a yearly basis, as well as the methodological choices inherent to efficiency modelling. Although extensive, these regularly published datasets from NHS Trusts could possibly contain inconsistencies in reporting, unobserved factors, or unquantified qualitative factors such as staff welfare or patient satisfaction. Additionally, the dataset published on the NHS Digital site is not real-time so dynamic allocation cannot be done. Thus, some particularly important parameters of healthcare performance, such as nuanced operational outcomes and financial attributes, are only indirectly addressed or might even be excluded altogether because systematic or standardized protocols of data collection have not been in existence.

Additionally, the process of selecting and weighting input and output variables for both DEA and MCDA frameworks can be done differently based on various priorities. Although sensitivity analyses were employed to mitigate the impact, these frameworks nonetheless rely on expert judgment and subjective determinations regarding which variables best capture organizational efficiency and performance. As a result, the conclusions drawn, especially scores and rankings, are likely to be dependent on such methodology assumptions, and therefore introduce some arbitrariness despite efforts to ensure robustness.

Future research may also consider temporal trends, unexpected external events (like the COVID-19 pandemic), or changing structural dynamics within trusts. However, the literature on effective application of behaviour combined with statistics in parallel to achieve measurable real-world outcomes in NHS settings is not sufficiently documented. In addition, the MCDA criteria were developed to fit the strategic purpose of the NHS, meaning that the framework may have some inapplicability in other foreign or differently structured health systems.

Future investigations in this field ought to emphasize the implementation of dynamic, longitudinal methods for efficiency analysis, particularly through the use of panel data DEA and MCDA models. This will allow for the monitoring of NHS Trust performance over time and separating the effects of organizational change, external policy interventions, and macroeconomic fluctuations, thus assisting with causal inference about which reforms bring about sustained improvements. It would also be particularly useful to apply behavioural principles in NHS Trusts, such as random control trials or realist evaluation, to test empirically the impact of different interventions such as feedback loops, recognition frameworks, or decision-oriented nudges on institutional efficiency and organizational culture.

Earlier advancements in modelling, particularly the combined use of efficiency frameworks (such as DEA with stochastic frontier Bayesian MCDA), can now be avenues along which prediction precision and dependability could still be strengthened. Beyond the methodological successes, stakeholder engagement-focused evaluations hold great promise: involving patients, clinicians, and Trust managers in the collaborative development and approval of assessment criteria will guarantee that the recommendations are practical and pertinent. To ensure that analytical frameworks are consistent

with NHS approaches to equity and inclusivity in resource allocation, the next step would be to research equity and access further using deprivation indices, geographic differences, and service accessibility measures used in efficiency studies.

6. Conclusion and Implications

This research aimed to concentrate on the effective allocation of healthcare resources by merging behavioural sciences with quantitative methods, particularly **Data Envelopment Analysis (DEA)** and **Multi-Criteria Decision Analysis (MCDA)**. The primary motivation arose from the acknowledgment that the quantitative precision of conventional efficiency models, although essential, frequently does not reflect the intricate realities of decision-making within the NHS influenced by cognitive biases, cultural inertia, and the experiences of stakeholders. The results and methodological innovations of this study emphasize the various possibilities inherent in integrating quantitative efficiency measures with a deeper behavioural insight.

Results demonstrated considerable differences in efficiency between NHS Trusts, with the **DEA indicating particularly significant differences** and emphasizing technical and scale inefficiencies. **The addition of MCDA then provided for much broader organizational outcomes, such as financial sustainability and operational scale**, which, when appropriately weighted, offer a more comprehensive perspective on Trust performance. The sensitivity analysis confirmed that while technical efficiency forms a solid foundation, real-world excellence requires alignment across multiple dimensions, as depicted in the strong but imperfect correlation between DEA and MCDA scores.

Importantly, the thesis demonstrated that effective resource allocation cannot rely solely on quantitative rankings. Decisions are made and policies are enacted by individuals operating within intricate organizations, influenced by motivational, cognitive, and environmental factors. By mapping efficiency outcomes to actionable recommendations, tailored by both behavioural science and empirical scoring, the thesis offered a practicable pathway for NHS administration to drive improvement. **For Trusts that excel, strategies centred on positive reinforcement can help maintain optimal practices.** In contrast, **those with moderate or low performance levels can gain from specific nudges, feedback mechanisms, and, when required, comprehensive structural changes.**

Moreover, **a strategy to combine and embed both behavioural insight and efficiency improvement in policy design, performance monitoring, and investment policy that allocates funds and resources** such as an NHS framework is clearly warranted. The reproducible process that stems from this documentation can be used for learning, accountability, and adaptability management, alongside regular benchmarking.

This research also points to key areas for further development. Some of the limitations that accompany cross-sectional analysis such as available data sources or transfer of behavioural intervention to tangible institutions outcomes indicate an urgent need for continued research. **Future studies should attempt to establish more dynamic, longitudinal models**, extend performance measures to equity, patient experience, and social value, and evaluate the impact of behavioural interventions using field experiments and participatory design. This would also bring in stakeholder input into developing a collaborative framework for resource allocation with a view toward further closing the long-standing gap between analytics and practice.

By highlighting the connections between quantitative performance and behavioural reality, this study highlights analytical resources for NHS resource planners and opens new vistas to more equitable, effective and sustainable healthcare delivery in the future. **The integrated use of DEA, MCDA, and behavioural sciences** establishes a persuasive framework for both academic research and significant policy reform, one that is equipped to address the challenges faced by modern healthcare systems.

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